

Announced Rescue Pricing with Latent Market Thickness

Group-meeting note — benchmark, numerical pattern, and formal model

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Preview. A publicly known single driver gives an exact cutoff-WPBE and a closed-form optimal menu. With latent Poisson supply, the completion value of rescue pricing is small in thin markets, largest at an intermediate thickness, and small again in thick markets. For $(\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0)$, the deterministic numerical solution peaks near $m = 9.72$ and improves optimized completion by 6.16 percentage points. The endpoint pattern is proved; strict single-peakedness is a conjecture.

1 Exact one-rider/one-driver benchmark

1.1 Closed-form solution

The rider has $v \sim U[0, 1]$, the driver has $c \sim U[0, 1]$, and the platform maximizes completion. Delayed rider value is βv and delayed incumbent surplus is $\delta(p_j - c)$. The platform commits to $0 \leq p_1 \leq p_2 \leq 1$. The number of drivers is publicly known to be exactly one; there is no survival shock or fresh entry.

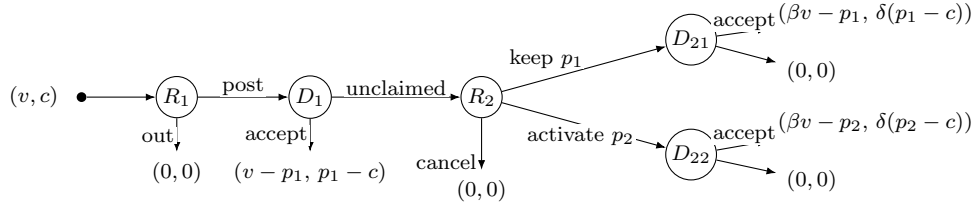


Figure 1: Extensive form.

For an active menu $0 \leq p_1 < p_2 < \beta$, the driver accepts immediately iff $c \leq \hat{c}$. After rejection, the rider cancels, repeats p_1 , or activates p_2 according to

$$v < \frac{p_1}{\beta}, \quad \frac{p_1}{\beta} \leq v \leq v^M(\hat{c}), \quad v > v^M(\hat{c}), \quad v^M(\hat{c}) = \frac{p_1 + p_2 - \hat{c}}{\beta}.$$

The marginal driver's accept-versus-wait indifference gives the WPBE cutoff

$$\hat{c}(p_1, p_2) = \left[p_1 - \frac{\delta(p_2 - p_1)(\beta - p_2)}{\beta(1 - p_1) - \delta(\beta - p_2)} \right]_+, \quad (1)$$

and equilibrium completion is

$$M_1(p_1, p_2) = (1 - p_1)\hat{c} + \frac{(\beta - p_2)(p_2 - \hat{c})}{\beta}. \quad (2)$$

Thus the cutoff is calculated from the WPBE; the platform then optimizes over menus subject to this equilibrium response.

Fixing a target cutoff reduces the two-price problem to one dimension. The unique continuation envelope and its implementing first price are

$$\hat{p}_2(\hat{c}) = \frac{\beta + \hat{c}}{2}, \quad \hat{p}_1(\hat{c}) = \frac{1 + \hat{c} - \sqrt{\Delta(\hat{c})}}{2}, \quad \Delta(\hat{c}) = (1 - \hat{c})^2 - \frac{\delta}{\beta}(\beta - \hat{c})^2. \quad (3)$$

Substitution into (2) gives

$$M_1^* = \max \left\{ \frac{1}{4}, \max_{\hat{c} \in [0, \beta]} J_\delta(\hat{c}) \right\}, \quad J_\delta(\hat{c}) = \frac{\hat{c}[1 - \hat{c} + \sqrt{\Delta(\hat{c})}]}{2} + \frac{(\beta - \hat{c})^2}{4\beta}. \quad (4)$$

If $\beta \leq 1/2$ or $\delta = 1$, the optimal value is the flat benchmark $1/4$. If $\beta > 1/2$ and $\delta < 1$, J_δ is strictly concave and has a unique interior maximizer, which implements $p_1^* < p_2^*$.